| INCH POUND |

MIL-T-23759D
28 February 1989
SUPERSEDING
MIL-T-23759C
26 March 1984

MILITARY SPECIFICATION

TANK, ASPHALT STORAGE, WHEEL MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers various size, commercial design, insulated, mobile, asphalt storage tank to be transported empty to a remote area and permanently set up as one component of a heavy-duty asphalt mixing plant.

1.2 Classification. The asphalt storage tank will be one of the following sizes, as specified (see 6.2).

1.2.1 Size. The size of tanks will consist of the following:

a. Size 1: 4,000 gallons.
b. Size 2: 5,000 gallons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5430

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SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation; Methods of.
MIL-T-704 - Treatment and paint of Material.
MIL-P-14105 - Paint, Heat-Resisting.
MIL-C-46168 - Coating, Aliphatic Plyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.

HANDBOOK

MIL-HDBK-113 - Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN WATER WORKS ASSOCIATION, INC. (AWWA)

AWWA D100 - Standard for Steel Tanks, Standpipes, Reservoirs, and Elevated Tanks, for Water Storage.

(Application for copies should be addresssed to the American Water Works Association, Inc., 6666 West Quincy Avenue, Denver, CO 80235.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The mobile asphalt storage tank shall consist of an insulated steel tank equipped with chassis, heating coils, and other equipment specified herein. The coils shall be designed for connection to an independent hot-oil heating source. The tank shall be mounted on a structural steel chassis equipped with a single axle suspension system with dual wheels or single wide base wheels and pneumatic tires. When prepared for travel, the mobile tank shall be compatible for coupling, towing, and maneuvering under all conditions specified herein.

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3.2 First article. When specified (see 6.2), the contractor shall furnish one complete mobile tank for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Material deterioration prevention and control for Army procurements. The mobile asphalt storage tank shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.2 Identification of materials and finishes for Army procurements. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Design and construction. The mobile tank shall be of all steel construction and shall be in accordance with the best commercial practice, as specified herein.

3.5.1 Dimensions and clearances. The empty mobile tank, uncoupled from the towing vehicle, resting level on the landing legs on level ground, shall conform to the dimensions and clearances specified in table I.

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TABLE I. Dimensions and clearances.

Overall width (not more than)	96 inches
Overall length	Minimum practicable (not more than 32 feet 10 inches, see figure 1)
Overall height (not more than)	144 inches
Ground clearance	Maximum practical but not less than 12 inches
Turning clearance radius from center of kingpin to nearest portion of chassis 6 inches or more below the upper fifth wheel plate (not less than)	82 inches
Upper fifth wheel height from ground (±1 inch)	55 inches
Swing radius from centerline of kingpin to most distant point on front end of chassis nose (not more than)	56 inches
Distance from centerline of kingpin to front end of chassis (not less than)	18 inches

3.5.1.1 Maintainability for Army procurements. The mobile tank shall conform to MIL-STD-1472, 5.9 (Design for Maintainability).

3.5.2 Chassis. The chassis components shall be adequate to support the loads imposed when the mobile tank is transported with the tank empty and when the mobile tank is supported on the landing legs at the front end and on blocking at the opposite end with the tank completely filled. Load bearing members shall not be subjected to a maximum tensile fiber stress greater than 50 percent (%) of the material yield strength when statically loaded with specified load herein and shall not result in any permanent sagging or warping of the structure. All parts of the chassis frame, where welding is required, shall be electric welded.

3.5.2.1 Upper fifth wheel plate. The upper fifth wheel plate shall be of the 1-piece rigid type, not less than 3/8-inch (in.) thick, reinforced and of sufficient size to cover a lower truck fifth wheel plate of 36 inches in diameter and shall conform to DoT Motor Carrier Safety Regulations, section 393.70(b). The kingpin shall be made of heat-treated alloy steel and shall conform to SAE J700. The forward end of the upper fifth wheel plate shall have a turned-up lip for ease of loading and for body protection.

3.5.2.2 Landing gear. Each chassis shall have two vertical, steel frame, retractable, landing legs capable of withstanding (without permanent distortion) the static loads imposed by the proportionate weight of the chassis and tank filled with asphalt. Each landing leg shall have a steel pad not less than 12-inches square (in^2). Edges of pads shall be turned up.

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Landing legs shall operate freely and shall be protected from malfunctioning caused by foreign matter. When placed into travel position, the landing legs shall remain positively locked to provide ground clearance in the course of normal towing operations.

3.5.2.3 Brakes. Brakes shall be of the full air type. The brakes shall conform to DoT, Federal Motor Vehicle Safety Standard No. 121 and Federal Motor Carrier Safety Regulations, sections 393.45 through 393.47, and as specified herein. The braking system shall include manual or automatic slack adjusters, piping, hose connections, glad hands, spring-loaded dust covers or dummy glad hands equipped with security chains, and all other components required for a complete system. Airhose fittings and location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims.

3.5.2.4 Suspension. The axle fo the mobile tank shall be a standard, commercial, one piece through shaft axle or tubular or structural shape steel without springs.

3.5.2.5 Wheels, tires, and tubes. Chassis for the mobile tank shall be equipped with commercial-type disc wheels, rims, tires, and tubes. Wheels and tire ratings shall conform to the TRA recommendations for the type and size tires furnished. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with tank empty. When tube type tires are furnished, inner tubes shall be of heavy-duty type and shall be of proper size. Tire flaps shall be provided for tube-type tires in accordance with TRA recommendations. Tires furnished to the Government shall be not more than 12 months old on the date of shipment by the supplier. Tire pressure, as recommended by the TRA year book shall be stenciled in one inch high letters "TP (specify) LB" above or near the tire. When specified (see 6.2), wide base wheels and tires shall be furnished.

3.5.2.6 Spare wheel assembly. An inflated tire, on spare wheel or rim, shall be furnished. The spare wheel assembly shall be mounted on a carrier and installed in a readily accessible location under chassis or body. Carrier shall permit removal or installation of spare assembly.

3.5.2.7 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.5.2.8 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. When such contact cannot be avoided, joints between dissimilar metals shall be

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protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fastenings shall be of the same material as the metals joined or shall be cathodic to the metals joined.

3.5.3 Tank. The tank shell shall be the manufacturer's standard shape having either a square bottom with rounded top, a round cross secion, or an oval shape; unless otherwise specified (see 6.2) the nominal capacity shall be 4,000 gallons (gal), plus a 5 percent allowance for expansion, when specified (see 6.2) the nominal capacity shall be 5,000 gals. The tank shall be fabricated of high strength, low alloy, hot-rolled steel plate of not less than 3/16-inch thickness. Necessary bulkheads, stiffeners, gussets, and baffles shall be installed.

3.5.3.1 Manhole. A manhole with filling cover shall be provided on the vertical centerline of tank at or near the longitudinal center and shall be manufacturer's standard commercial product. Manhole shall be not less than 12 x 16 inches. Manhole shall be fitted with a 10-inch fill cover with gaskets and locking clamps to provide a leakproof closure but shall operate freely by hand.

3.5.3.2 Heating coils. The tank heating coils shall be double, bank manifold for split flow with staggered tiers and shall be fabricated of 2-inch standard steel pipe or 2-inch commercial grade steel tubing. All coil sections inside the tank area shall have butt welded joints. The tank shall have not less than 240 lineal feet of heating coils for the 4,000 gal capacity size or not less than 300 lineal feet of heating coils for the 5,000 gal capacity size.

3.5.3.3 Fittings and connections. All valves shall be attached to the tank by means of a nipple welded at one end to the tank and threaded at the other to receive the valve. The nipple shall be long enough to provide adequate clearance from the tank for manual turning of the valve wheel. Welded and threaded connections shall be as specified herein. All valves shall be operable at temperatures to 400 degree (°) Fahrenheit (F). The mobile tank shall have a hot-oil inlet connection with a 2-inch steel, screwed, 125 psi, gate valve and a hot-asphalt outlet connection with a 3-inch steel, screwed, 125 psi, gate valve shall be furnished on the front (towing end) of the tank. An asphalt inlet connection located near the top of the tank with a 3-inch steel, screwed, 125 psi, gate valve and a drain connection at the bottom of the tank with a 3-inch steel, 125 psi, gate valve shall be furnished on the back end of the tank. In addition, a 3-inch male ground joint, 125-psi fitting shall be provided on the hot-asphalt outlet and on the drain connection.

3.5.3.4 Catwalk. A catwalk of the open grating type shall extend the full length of the top of the tank. The catwalk shall be integral with or securely attached to the tank.

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3.5.3.5 Ladder. A ladder of tubular construction, or equal, shall be securely installed at the rear of the tank to provide ready access to the cat-walk. Ladder rails shall not be used to hold wiring conduits or drain tubes.

3.5.3.6 Thermometer. A thermometer shall be installed on the tank so that the average temperature of the material in the tank can easily be read by operating personnel standing on the ground. The thermometer shall have a range of 50°F to 400°F and shall be liquid in glass type or dial type.

3.5.3.7 Insulation. The exterior of the tank shell shall be insulated with a fiberous insulating material not less than 2 inches in thickness and shall have a mimimum K-factor rating of 0.33°F at 150°F per inch of thickness. The insulating material shall be installed in such a manner as to prevent packing or shifting during use or travel. The insulation shall be protected by a weathertight sheath constructed of not less than No. 16 gage sheet steel or 0.050-inch aluminum attached over the tank so as to exclude moisture.

3.5.4 Electrical systems.

3.5.4.1 Lighting systems.

3.5.4.1.1 Basic lighting system. The lighting system shall be the manufacturer's standard, 12-volt (V) direct current, potential system and shall comply with DoT Federal Motor Carrier Safety Regulations, sections 393.14; 393.20; 393.25; 393.26 (a), (b), and (c); 393.27 through 393.29; 393.32; and 393.33. The system shall include at least two of each of the following: stoplights and tail-lights, directional signals, amber front and red rear marker, clearance lights, and reflector devices. The front of the mobile tank shall be equipped with a receptacle conforming to SAE J560 with the receptacle located and its conductors connected and color coded as specified therein. Lights and reflectors shall be recessed or guarded to protect them from damage during normal operation of the mobile tank. All wiring shall be properly routed and anchored to protect against snagging or tearing away.

3.5.4.2 Reflectors. Reflectors shall be mounted in conformance to DoT Federal Motor Carrier Safety Regulations, section 393.26. In addition, two amber reflectors shall be mounted on the front corners of the vehicle at approximately the same height as those mounted on the sides.

3.6 Performance characteristics.

3.6.1 Turning ability. The mobile tank, fully equipped but not loaded, shall be capable of making complete turns to the right or left with the mobile tank assuming a 90° angle with the towing vehicle when tested in accordance with 4.4.1.

3.6.2 Braking ability. Service brakes shall conform to (DoT) Federal Motor Vehicle Safety Standard No. 121 when tested in accordance with 4.4.2.

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3.6.3 Chassis load. The mobile tank chassis frame, fully equipped, shall be capable of withstanding a static load equivalent to the weight of the tank when completely filled with asphalt, when tested as specified in 4.4.3.

3.6.4 Towing speeds and tracking ability. The mobile tank shall conform to the following towing and tracking criteria when tested as specified in 4.4.4.

3.6.4.1 Highway operation. The mobile tank shall be towed empty over-smooth, hard surfaces, primary highways at speeds up to 35 miles per hour (mph) in accordance with 4.4.4.

3.6.4.2 Trailing ability. The mobile tank shall trail the truck tractor without weaving, sideswaying, cramping, or bottoming when tested in accordance with 4.4.4. The mobile tank shall follow the truck tractor without exceeding 75% of the tracking deviation as specified in DoT Motor Carrier Safety Regulations, section 393.70(a).

3.6.4.3 Cross-country. The mobile tank shall negotiate improved and unimproved roads and open, rolling, hilly cross-country terrain at speeds up to 10 mph minimum when towed by the truck tractor in accordance with 4.4.4.

3.6.5 Leakage. The tank and heating coils shall not leak or rupture when tested in accordance with 4.4.5.

3.7 Transportability.

3.7.1 Railroad transportability. When specified (see 6.2) sectionalization for railroad transportability shall be within the dimensional limits shown on figure 1. Sectionalization and reassembly shall each be accomplished, without cutting or welding, in not more than 36 man-hours.

3.7.2 Landing ship troop (LST), transportability. The complete mobile tank shall be capable of being loaded aboard an LST in accordance with the dimensions shown in figure 2.

3.7.3 Lifting and tiedown attachments. When specified (see 6.2), the tank shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or III of MIL-STD-209. For Army procurements, lifting and tiedown attachments shall conform to type II of MIL-STD-209. Slinging provisions may be class 1 or 3; tiedown provisions class 2 or 3. A nonferrous transportation plate shall be provided and mechanically attached to the tank. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the tank on the carrier when shipped.

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3.8 Instruction plates. The mobile tank shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous metal screws or bolts of not less than 1/8-inch diameter.

3.9 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous metal screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates. Nomenclature shall be "TANK, ASPHALT STORAGE, WHEEL MOUNTED, 4,000 GALLON CAPACITY," 5000 gallon capacity when applicable which either is specified (see 6.2).

3.10 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.10.1 Treatment and painting for Army procurements. The portions of the mobile tank normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168. The mobile tank shall then be overcoated in accordance with the Government furnished camouflage pattern (see 6.3) with MIL-C-46168 top coat of the colors specified in the camouflage patterns with the exception that the first article may be camouflage green 383. Heat resistant paint conforming to MIL-P-14105 shall be used on parts of the mobile asphalt storage tank subjected to a temperature of 400°F or higher. Unless otherwise specified, color per MIL-P-14105 shall be in accordance with FED-STD-595, lusterless dark green number 34079. As soon as practicable after cleaning and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated with one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall

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be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 miles over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects."

3.10.2 Government-furnished property. Unless otherwise specified, the following property as applicable in the quantities indicated will be furnished by the Government (see 6.2).

Description	Identification	Quantity
Pattern drawings	Camouflage patterns	As required

3.11 Identification marking. When specified (see 6.2), identification shall be permanently and legibly marked directly on the tank or frame or on a corrosion-resisting metal plate securely attached to the tank or frame at the source of manufacturer. Identification shall include the manufacturer's model and serial numbers, name, and trademark to be readily identifiable to the manufacturer.

3.12 Workmanship.

3.12.2 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.12.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.12.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.12.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.13 Safety. All exposed rotating or reciprocating parts, electrically energized parts and all parts subject to high operational temperatures that are of such a nature or so located as to be or become a hazard to the operating or maintenance personnel shall be insulated, envlosed, or guarded to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions. The tank and trailer shall comply with all applicable requirements of DoT carrier and OSHA safety regulations in effect at time of manufacture. Steel walkways and ladders shall be provided where necessary for inspection, servicing, and maintenance. All walkways, platforms, steps, and ladder rungs shall have anti-skid surfaces. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required and where they are not detrimental to safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one mobile tank when a first article is required (see 3.2, 6.2, and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a

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standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each mobile tank shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. The tank shall be examined as specified herein for the following defects:

CRITICAL

1.	Camouflage pattern data not as specified.	3.10

MAJOR

101.	Materials not as specified.	3.3
102.	Interchangeability not as specified.	3.4
103.	Design and construction not as specified.	3.5
104.	Dimensions and clearances not as specified.	3.5.1
105.	Chassis not as specified.	3.5.2
106.	Upper fifth wheel plate not as specified.	3.5.2.1
107.	Landing gear not as specified.	3.5.2.2
108.	Suspension not as specified.	3.5.2.4
109.	Wheels, rims, tires, and tubes not as specified.	3.5.2.5
110.	Spare wheel assembly not as specified.	3.5.2.6
111.	Lubrication not as specified.	3.5.2.7
112.	Tank not as specified.	3.5.3
113.	Manhole not as specified.	3.5.3.1
114.	Heating coils not as specified.	3.5.3.2
115.	Fittings and connections not as specified.	3.5.3.3
116.	Catwalk not as specified.	3.5.3.4
117.	Ladder not as specified.	3.5.3.5
118.	Insulation not as specified.	3.5.3.7
119.	Lighting system not as specified.	3.5.4.1
120.	Lifting and tiedown attachments not as specified.	3.7.3
121.	Workmanship not as specified.	3.13
122.	Safety not as specified.	3.14

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MINOR

201.	Reflectors not as specified.	3.5.4.3
202.	Instruction plates not as specified.	3.8
203.	Identification plate not as specified.	3.9
204.	Cleaning, treatment, and painting not as specified.	3.10
205.	Color not as specified.	3.10
207.	Camouflage pattern not as specified.	3.10
208.	Identification marking not as specified.	3.11

4.4 Tests. The first article mobile tank shall be subjected to the tests specified in 4.4.1 through 4.4.5. Each subsequent production mobile tank shall be tested for leaks in accordance with 4.4.5. Failure of any mobile tank to meet one or more requirements of the specified tests shall be cause for rejection.

4.4.1 Turning ability test. The mobile tank, when connected to the towing unit, shall be operated through 10 complete operational cycles turning 90° to the left and right. On completion or during this test, the equipment shall not show evidence of abnormal binding, permanent distortion, or other defects (see 3.6.1).

4.4.2 Service brake test. The mobile tank fully equipped, but not loaded, and connected to the towing unit shall be driven on smooth, dry, level pavement and brakes applied from a speed of 20 mph to determine conformance to the stopping distance required under 3.6.2. Breakaway situation shall be simulated to verify braking ability upon breakaway.

4.4.3 Chassis load test. The mobile tank, fully equipped, with axle and blocked up to clear tires and with landing gear set in operating position, shall be filled with water. There shall be no permanent deformation or damage to structural members or landing gear under this load condition (see 3.6.3).

4.4.4 Towing speeds and tracking ability test. Tow the empty mobile tank using a commercially rated 5-ton truck tractor over highways for not less than 100 miles at variable speeds up to 35 mph. Tow the empty tank over unimproved (unpaved) roads for not less than 50 miles at variable speeds up to 10 mph. Tow the empty tank over rolling hilly, cross-country terrain for not less than 25 miles. During or after the cumulative 175 miles of roading, trailing, and turning, the braking ability shall be observed. Inability of any trailer to operate, trail, turn, retard, and stop as specified in 3.6.4 or damage to or permanent deformation of any part of the tank shall constitute failure of this test.

4.4.5 Leakage test. After all welding is completed on the tank shell and heating coils are installed, each tank shall be tested at an internal pressure of 5 psi or by vacuum testing method specified in AWWA D100. The heating coils shall be subjected to a hydrostatic test of not less than 275 psi, maintained for a period of 15 minutes. Any failure or leaks shall be cause for rejection (see 3.6.5).

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4.5 Test procedures of lifting provisions. When specified (see 6.2) the following lifting tests shall be conducted:

a. Lift the mobile tank and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the mobile tank. Determine the forward angles of application for each provision. Measure the clearance between each sling and the mobile tank. Measure eye openings and clearance dimensions of the provision. Measure the load carried by each provision.

b. Restrain the mobile tank by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in a. above. An alternate method of test is to lift the mobile tank as in a. above and add weights to the main frame until required force is obtained. Hold the mobile tank under either of the above loading conditions for at least 90 seconds. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.5.1 Test procedures of tiedown provisions. When specified (see 6.2), the following test shall be performed. With the mobile tank anchored by means other than the provisions being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown. Failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs as listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

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5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1.

5.1.1.4 Tank. The interior of the asphalt storage tank shall be preserved with type P-10, grade 30 preservative.

5.1.1.5 Trailer chassis. The complete trailer chassis shall be preserved in accordance with the requirements of MIL-STD-281.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. The unit shall be shipped mobile. The unit shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations. The unit shall not be shipped resting on its landing gear.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mobile asphalt storage tank is one component of a heavy-duty, asphalt-aggregate mixing plant for advance base use. The tank will be empty when transported.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size classification of tanks (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required for examination and test (see 3.2, 4.2.1, and 6.4).
e. When wide base wheels and tires are acquired (see 3.5.2.4).
f. When lubrication, other than manufactures standard, is required (see 3.5.2.7).
g. When the tank shell should be the manufacturers standard shape (see 3.5.3).

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h. When the normal capacity shall be 4000 gallons plus 5 percent allowance for expansion (see 3.5.3).
i. When sectionalization for railroad transportability is required (see 3.7.1).
j. When lifting and tiedown attachments are required (see 3.7.3).
k. When tank capacity required is other than 4000 gallons for identification plate nomenclature (see 3.9).
l. When cleaning, treatment, and painting should be other than specified (see 3.10).
m. Color of finish paint required (see 3.10 and 3.10.1).
n. When identification marking is required (see 3.11).
o. When camouflage pattern data not as specified (see 3.10.1.3).
p. When test procedures of lifting provisions are required (see 4.5).
q. When test procedures of tiedown provisions are required (see 4.5.1).
r. Level of preservation required (see 5.1).

6.3 Data for camouflage patterns. The contractor will generate the following data:

a. Eight by ten color photos that mirror each of the views listed under 6.3.1.

b. Original illustrations of each of the views listed under 6.3.1. The illustrations shall be on D size paper. Each illustration shall include length, width, and height dimensions relative to each other and shall be clearly detailed to the extent that all discontinuities of the spreader that cover one square inch or more of area, are clearly delinerated to scale. Use 1/8 scale for an item with an overall length of 25 feet or less and 1/10 scale if the item is more than 25 feet.

6.3.1 Camouflage pattern views.

a. Back.
b. Front.
c. Left side.
d. Right side.
e. Top.
f. Hidden views (see 6.6) that would result in a change of a finished pattern.

6.4 Consideration of Data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the

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requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.10.1	DI-ADMIN-80424	Camouflage pattern photos	
3.10.2	DI-M-30420	Camouflage pattern illustrations	

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.5 First article. When a first article inspection is required (see 3.2, 4.2.1, and 6.2), the mobile tank will be tested and should be a first production item or it may be a standard production item or it may be a standard production item from the contractor's current inventory. The first article should consist of one complete mobile tank. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.10.1.3.

6.7 Hidden views. Hidden views are those areas requiring camouflage coloration that are not apparent from other views. Included are variations in the appearance of the spreader that result when the deployed spreader is taken from a stand-by condition and placed into a ready-for-use condition.

6.8 Lubricants for Army procurements. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research & Development Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under chapter 2 of MIL-HDBK-113.

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6.9 Subject term (key word) listing.

Mixing plant
Mobile
Asphalt
Storage

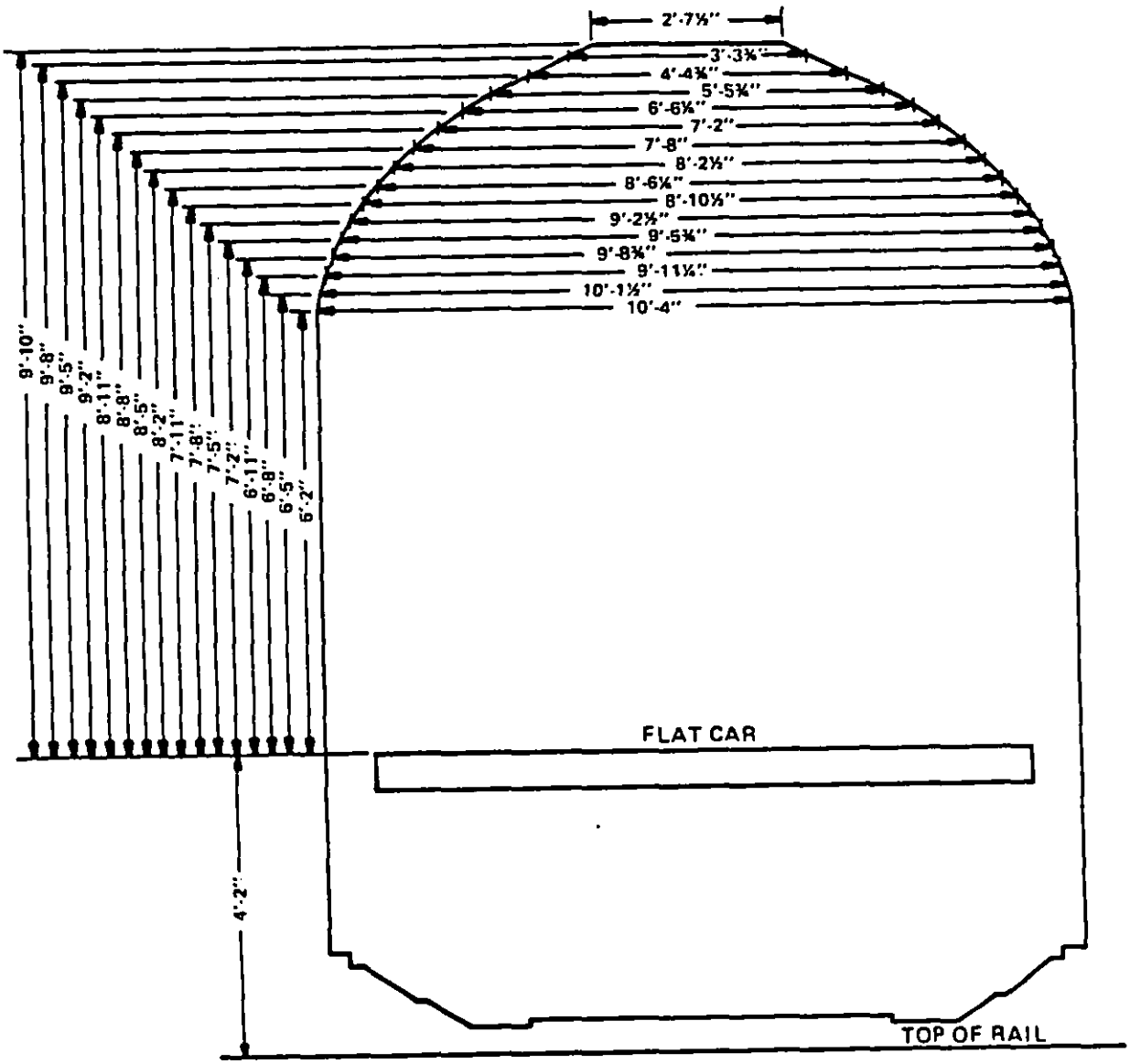
6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Preparing activity:
Navy - YD

(Project No. 5430-0188)

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MAXIMUM LOADING LENGTH - 32'-10"

FIGURE 1. Berne international (clearance diagram).

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LEGEND

Bow Ramp A

Capacity - 75 ton
Length - 100'
Width - 15'
Max ramp angle - 21
Clearance to overhead - Not applicable

Between Deck Ramp B

Capacity - 75 ton
Clearance to overhead:
Ramp down - 13'3"
Ramp up - 13'7"
Width - 15'8"
Length - 47'9"
Ramp angle - 19°

Deck Ramp

Bottom - 20'
Top - 10'
Height - 1'

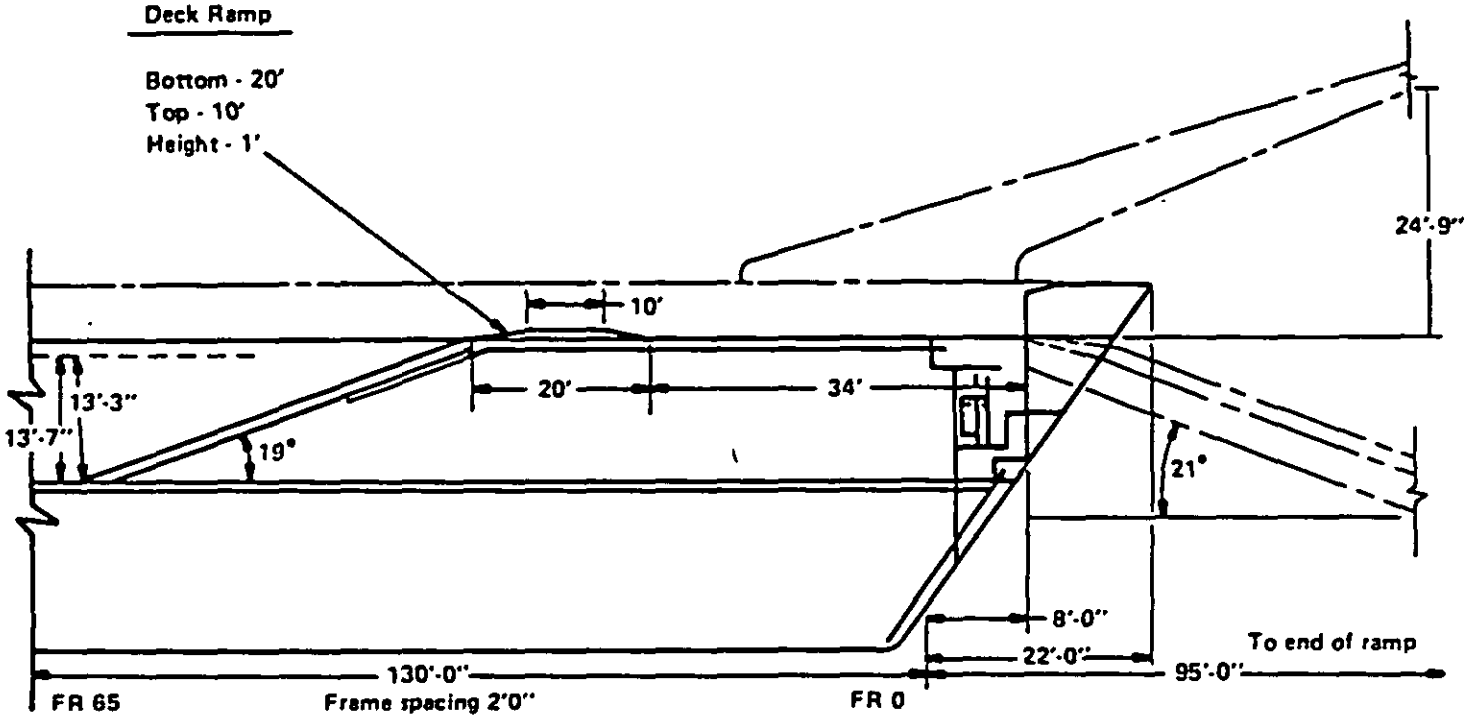


FIGURE 2. LST clearance.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-23759D	MILITARY SPECIFICATION _ TANK, ASPHALT STORAGE, WHEEL MOUNTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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